

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 5

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Recap of Previous Lecture



Topic

Direct Mapping



Topics to be Covered



Topic

Cache Initialization

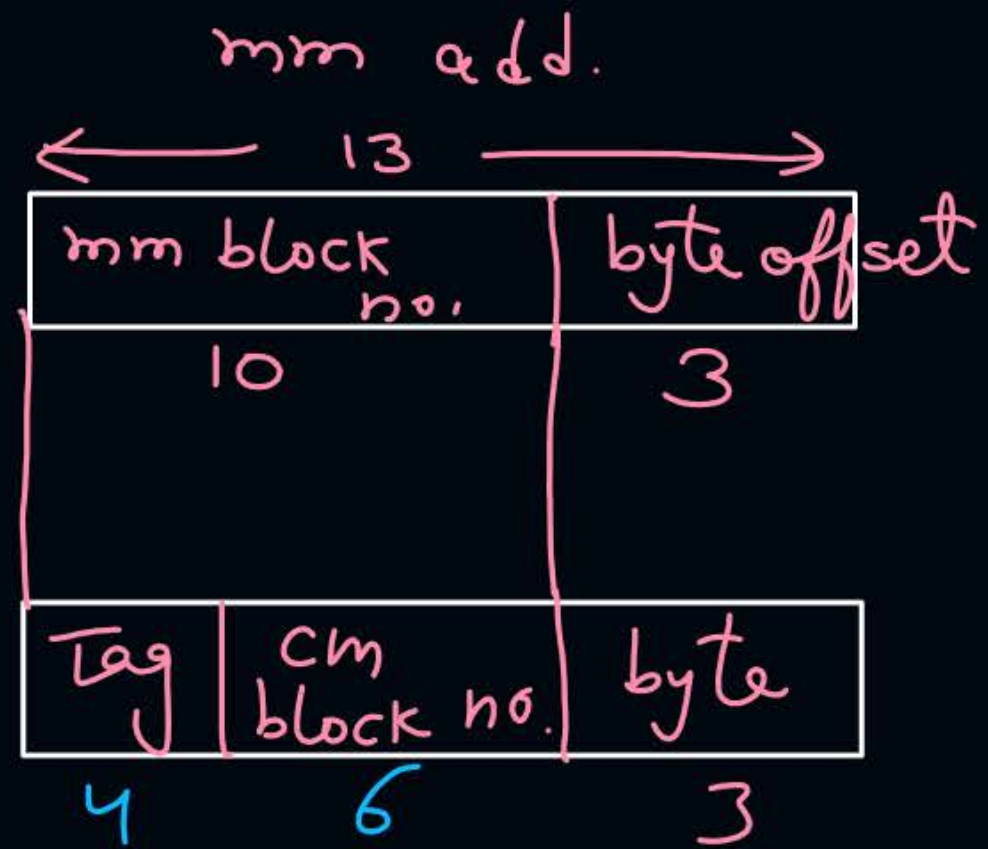
Topic

Set Associative Mapping

#Q. Consider a direct mapped cache of size 512B with block size 8 Bytes. The CPU generates 13-bits addresses.

1. Number of bits for byte offset? 3 bits
2. Number of blocks in cache? $2^6 = 64$
3. The number of bits needed for cache indexing? 6 bits
4. The number of tag bits? 4 bits
5. Tag Directory size? $2^6 * 4 \text{ bits} = 256 \text{ bits}$

2^3 B
↓
byte offset = 3 bits



$$\begin{aligned}
 \text{no. of blocks in cache} &= \frac{\text{Cache size}}{\text{block size}} \\
 &= \frac{512 \text{ B}}{8 \text{ B}} = \frac{2^9}{2^3} = 2^6 \Rightarrow \text{cm block no} = 6 \text{ bits}
 \end{aligned}$$

$2^4 \text{ B} \Rightarrow \text{byte} = 4 \text{ bits}$
 $\uparrow$
 offset

#Q. Consider a direct mapped cache of size 64KB with block size 16 Bytes. The CPU generates 32-bits addresses.

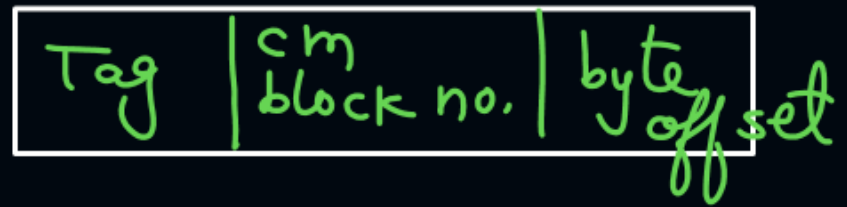
1. Number of bits for byte offset? 4 bits
2. Number of blocks in cache? 2^{12}
3. The number of bits needed for cache indexing? 12 bits
4. The number of tag bits? 16 bits
5. Tag Directory size? $2^{12} * 16 \text{ bits}$



$$\text{no. of blocks in cache} = \frac{64 \text{ KB}}{16 \text{ B}}$$

$$= 4 \text{ K} = 2^{12} \Rightarrow \text{cm block no.} = 12 \text{ bits}$$

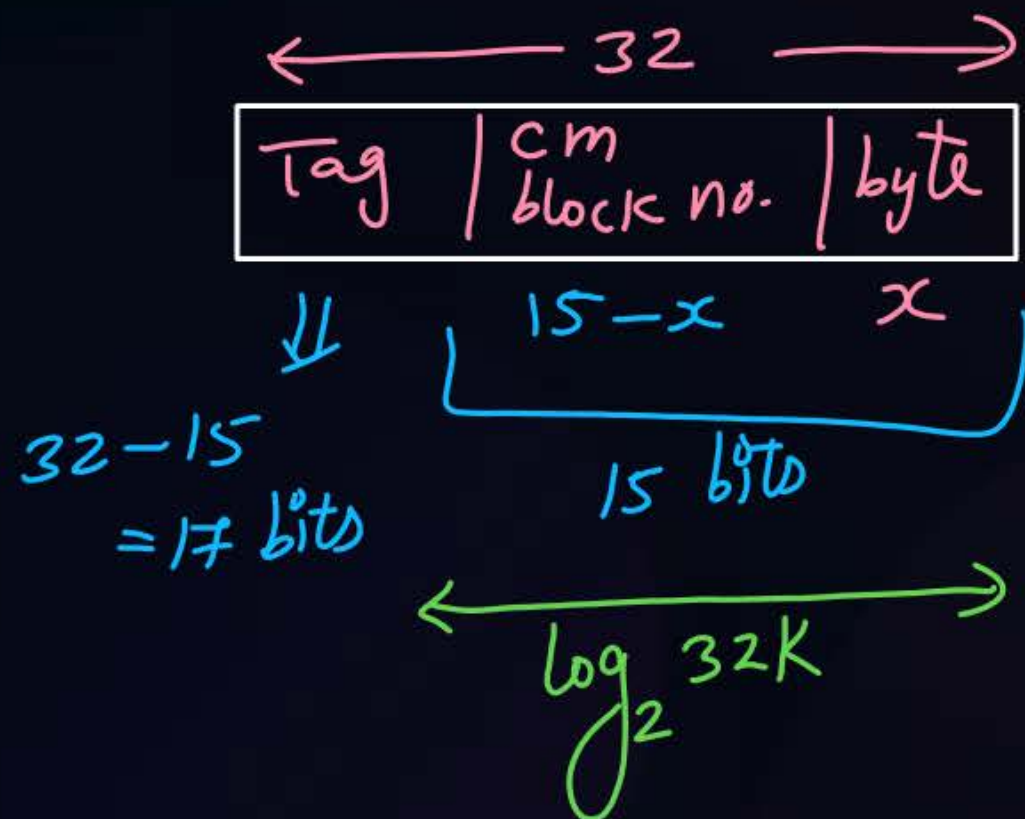
mm add.



$\leftarrow \log_2(\text{cm size}) \rightarrow$

#Q. Consider a direct mapped cache of size 32KB. The CPU generates 32-bits addresses. The number of tag bits in main memory address are?

block size = 2^x bytes $\Rightarrow$ byte offset = x bits



$$\text{no. of blocks in cache} = \frac{32K}{2^x} = \frac{2^{15}}{2^x} = 2^{15-x}$$

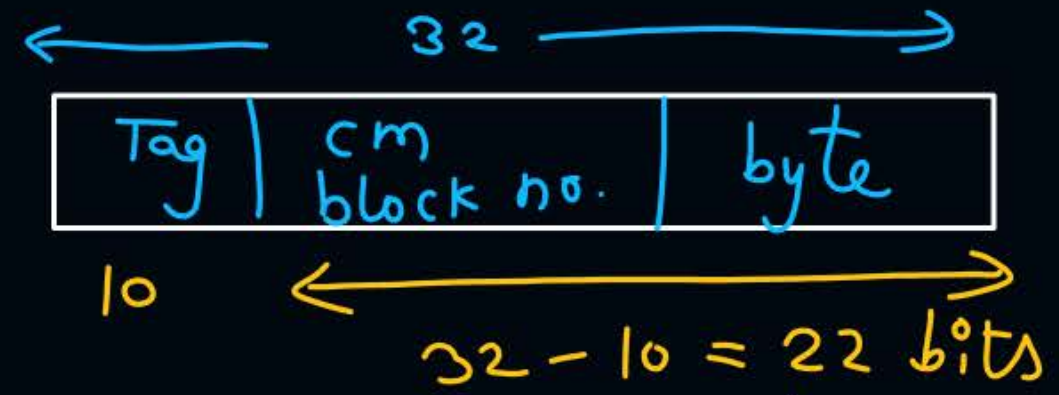
$\downarrow$

cm block no. = 15 - x bits

ques) Direct mapped cache
mem add. = 32 bits

Tag = 10 bits

cm size = 4MB ?



$$\begin{aligned} \text{cm size} &= 2^{22} \text{ bytes} \\ &= 4 \text{ MB} \end{aligned}$$



Topic : Cache Initialization

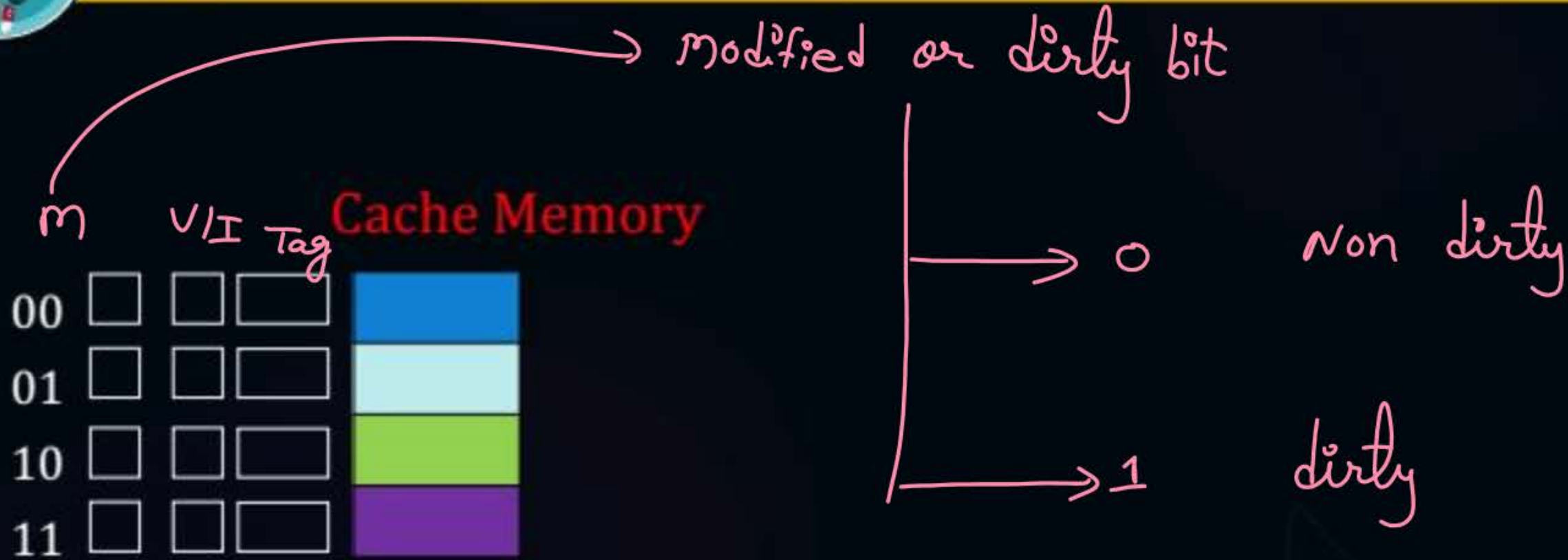
valid invalid bit { 0
1

block is invalid
block is valid

Cache Memory		
V/I	Tag	
00	☐	
01	☒	✓
10	☐	
11	☐	



Topic : Performance Improvement of Write Back Cache



$$\text{cache tag directory size} = \underset{\text{in cache}}{\text{no. of blocks}} * (\text{Tag} + \text{extra bits})$$

#Q. Block Size = 16 bytes $\Rightarrow$ byte offset = 4 bits
 Size of Cache memory = 128KB = 2^{17} B
 Size of Main memory address = 34-bits

Direct Mapping

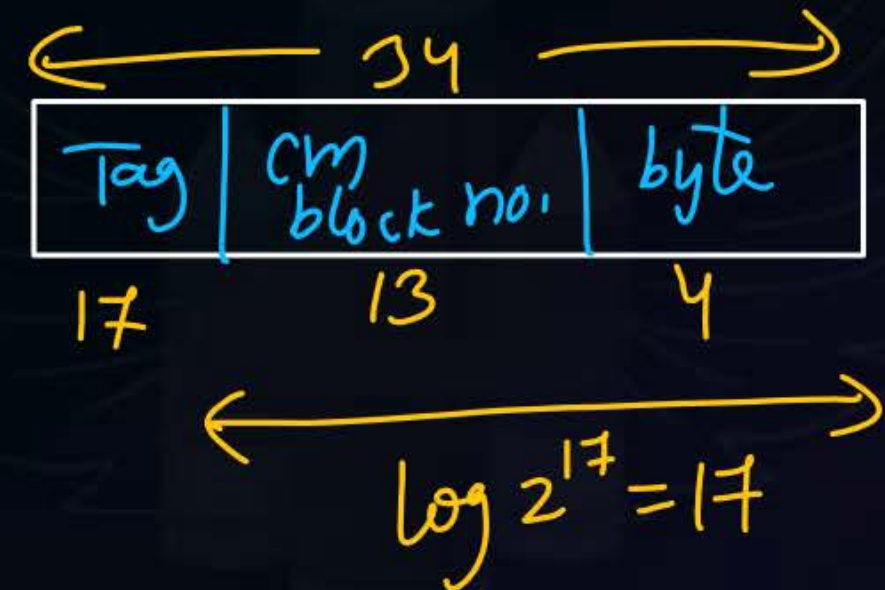
For each block apart from tag 1 valid bit and 1 modified bit are stored in cache

Bits in byte offset? 4 bits

Bits in cache block number? 13 bits

Bits in tag? 17 bits

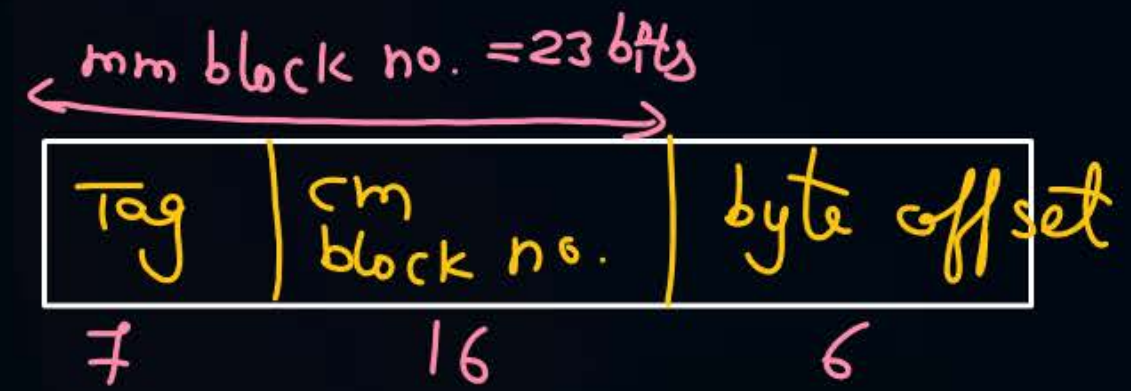
Tag Directory size? $2^{13} * (17 + 2) \text{ bits} = 2^{13} * 19 \text{ bits}$



[NAT]



#Q. Blocks in Main memory = 2^{23}
Blocks in Cache memory = 2^{16}
Block Size: 64 Bytes = 2^6 B
Direct Mapping



No. of bits required for Byte Offset = ? 6

No of bits required for main memory address = ? 23 bits

Index-bits = ? 16

Tag-bits = ? 7

Size of Tag Directory = ? $2^{16} * 7$ bits

#Q. 32-bit architecture CPU

Main Memory Size = 4GB = $2^{32} \text{ B} \Rightarrow \text{maddr.} = 32 \text{ bits}$

Cache Size = 256KB

Block Size = 16 Words = $16 * 4 \text{ B} = 64 \text{ B}$

Direct Mapping

No. of bits required for Byte Offset = ? 6

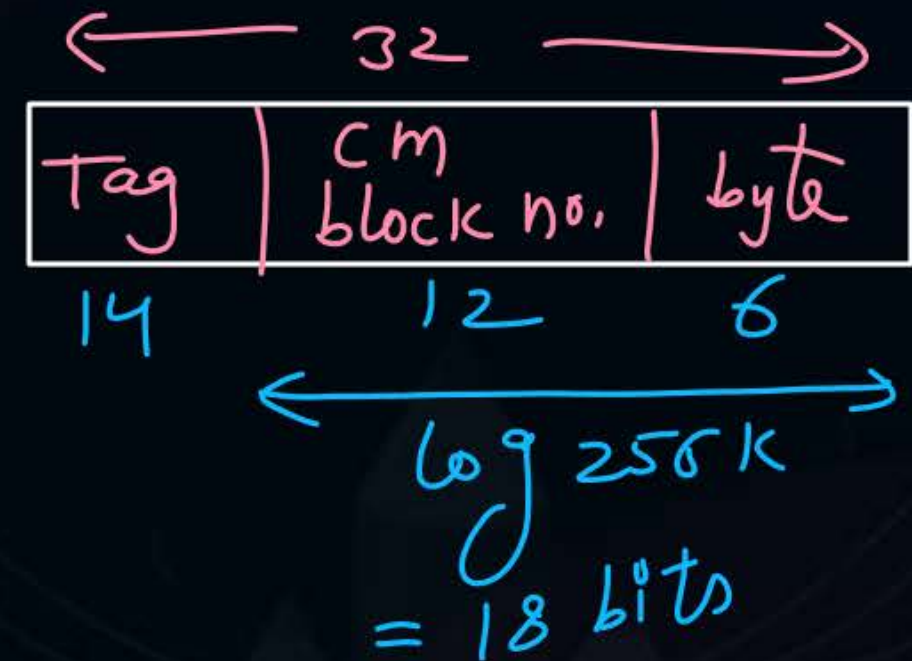
No of bits required for main memory address = ? 32

No of bits required for main memory block no. = ? 26

Index-bits = ? 12

Tag-bits = ? 14

Size of Tag Directory = ? $2^{12} * 14 \text{ bits}$





Topic : Calculating CM Block Number from MM Address

mm add. given

→ In binary

→ Hexadecimal

Tag	cm block no.	byte
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mm add. given in decimal:-

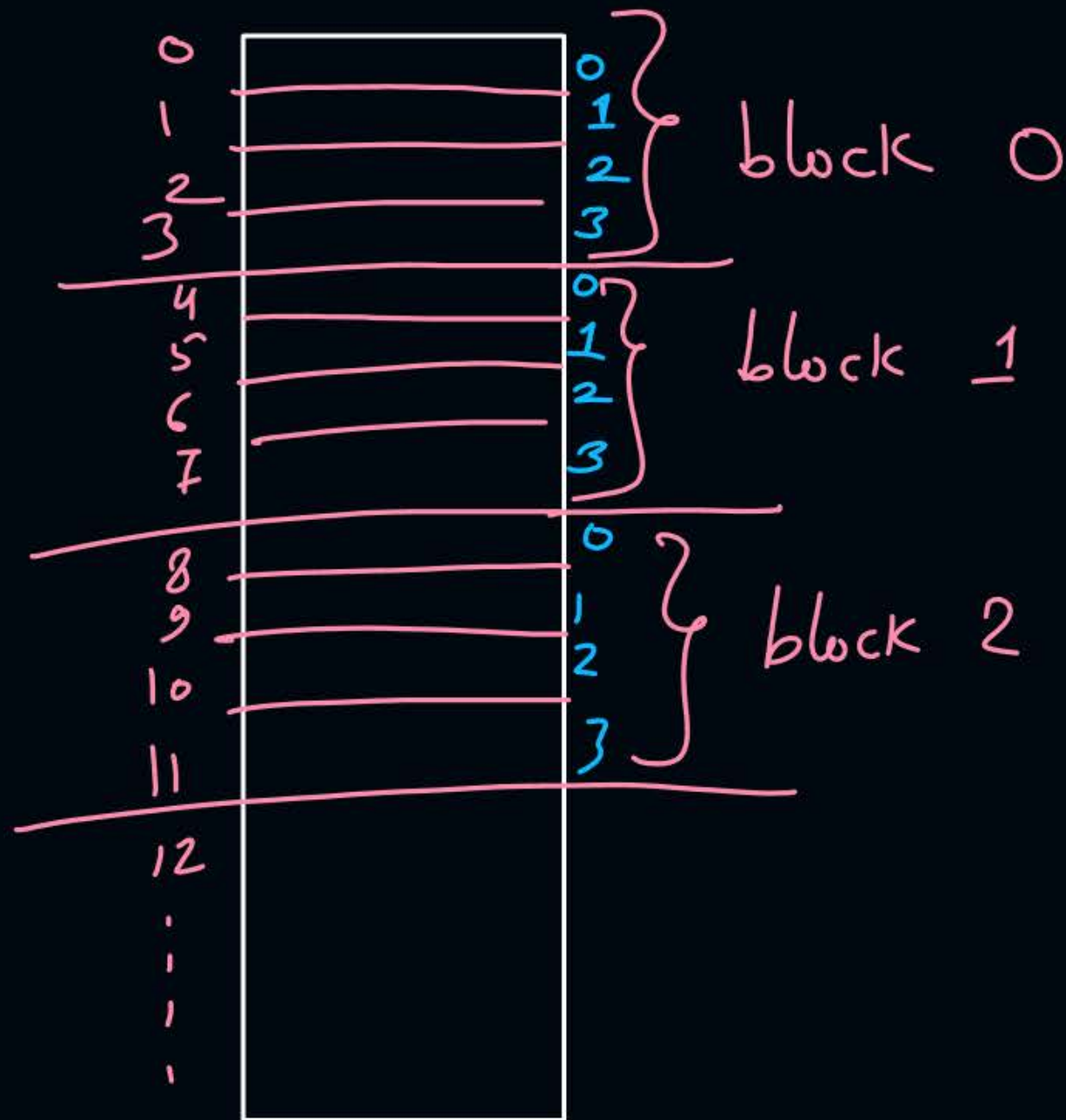
$$\text{mm block no.} = \left\lfloor \frac{\text{mm address}}{\text{block size}} \right\rfloor$$

$$\text{byte offset} = \text{mm address} \% \text{block size}$$

$$\text{cm block no.} = (\text{mm block no.}) \% \text{no. of blocks in cache}$$

$$\text{Tag} = \left\lfloor \frac{\text{mm block no.}}{\text{no. of blocks in cache}} \right\rfloor$$

ex:- block size = 4 bytes



Ques) mm address size = 16 bits

block size = 8 bytes

cm size = 128 bytes

$$\text{blocks in cache} = \frac{128}{8} = 16$$



mm add.	mm block no.	cm block no.	Tag
$(1123)_{10}$	$\left\lfloor \frac{1123}{8} \right\rfloor = 140$	$140 \% 16 = 12$	$\left\lfloor \frac{140}{16} \right\rfloor = 8$
$00 01 0000 0000$		$(6110)_2 = (6)_{10}$	$(001011000)_2$
Tag	block		
	byte		

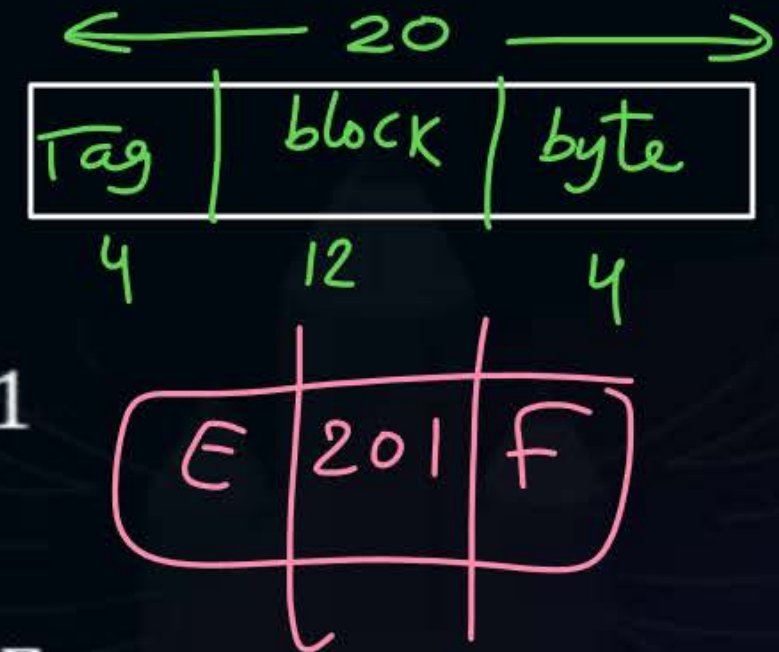
#Q. Consider a machine with a byte addressable main memory of 2^{20} bytes, block size of 16 bytes and a direct mapped cache having 2^{12} cache lines. Let the addresses of two consecutive bytes in main memory be $(E201F)_{16}$ and $(E2020)_{16}$. What are the tag and cache line address (in hex) for main memory address $(E201F)_{16}$?

A ✓ E, 201

C E, E20

B F, 201

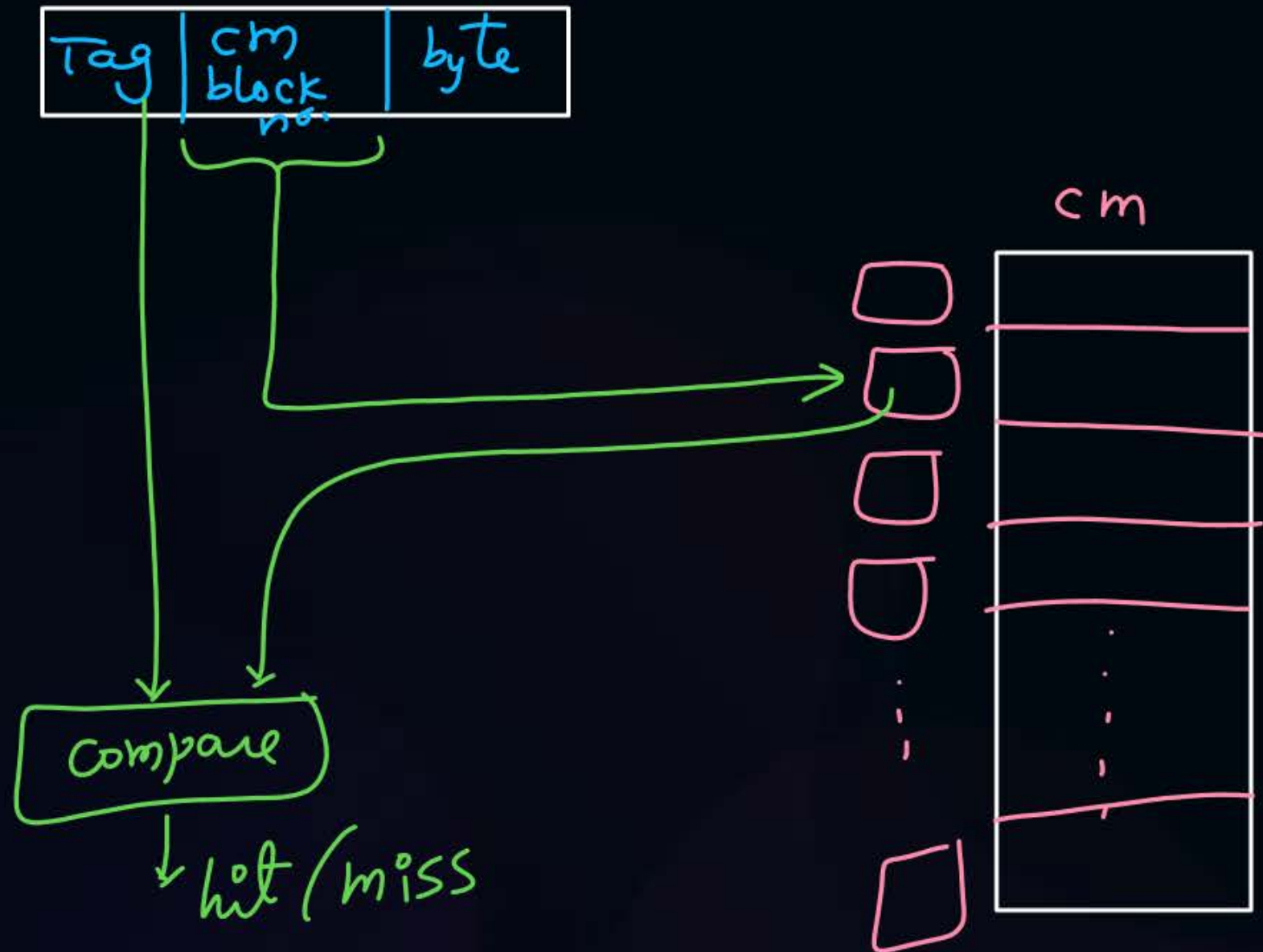
D 2, 01F





Topic : Checking Hit/Miss in Direct Mapped Cache

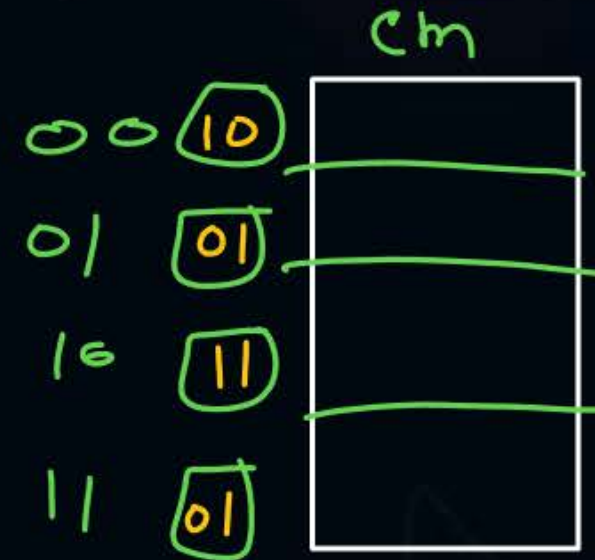
mem addr.



H. W.

#Q. Consider a 64 bytes direct mapped cache with a block size of 16 bytes. Main memory size is 256bytes. Currently in the cache, the blocks are having tags as follows:

Block	Tag
00	10
01	01
10	11
11	01

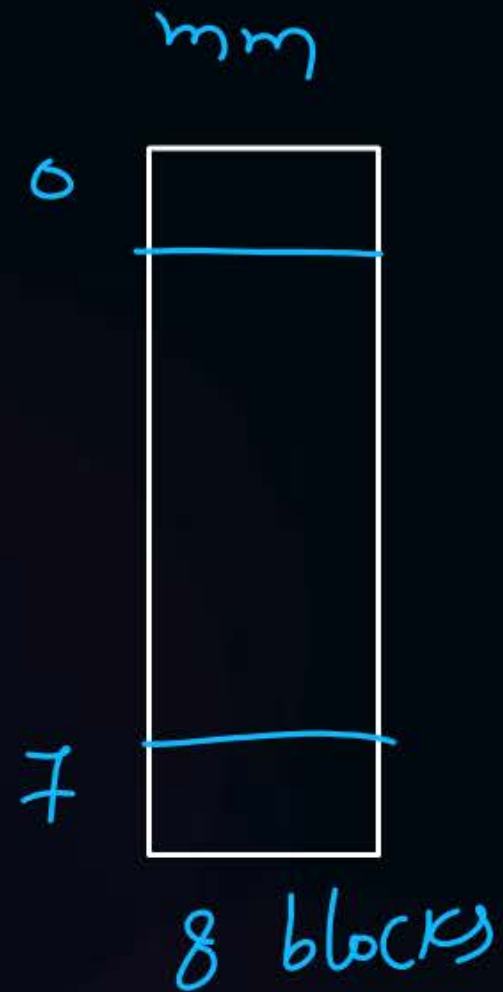
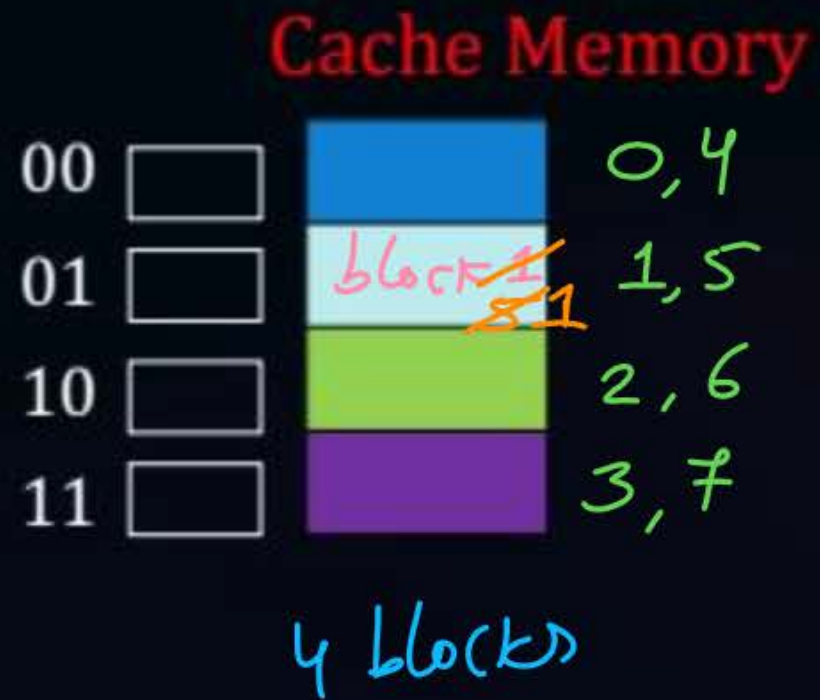


Identify the correct statement with respect to the availability of the main memory data into cache?

- a) Main memory byte number 243 present in cache
- b) Main memory byte number 143 present in cache
- c) Main memory byte number 43 present in cache
- d) Main memory byte number 119 present in cache

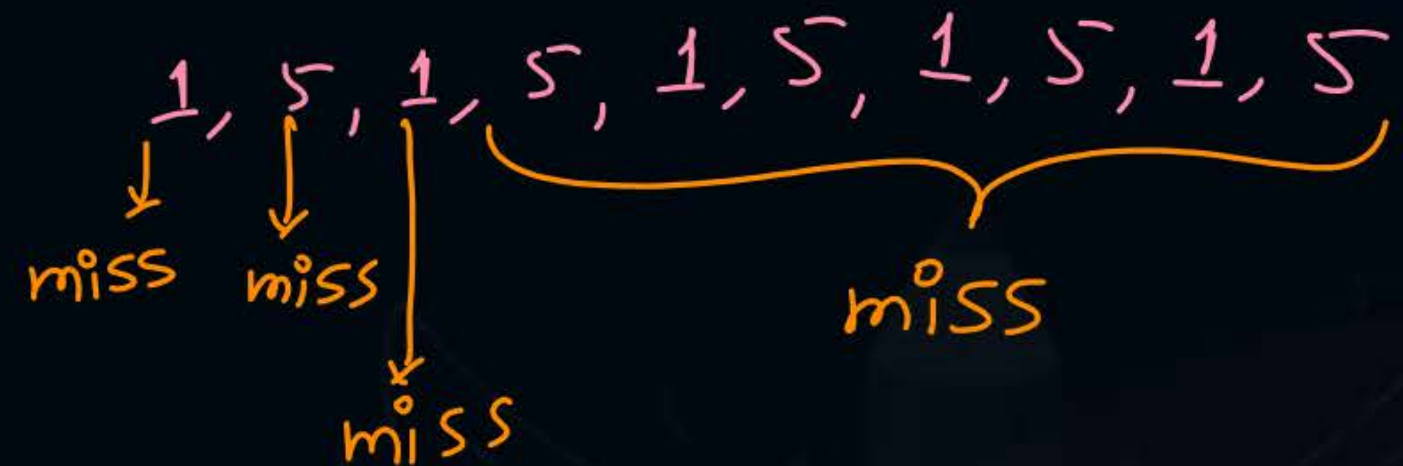


Topic : Problem With Direct Mapping



CPU requests

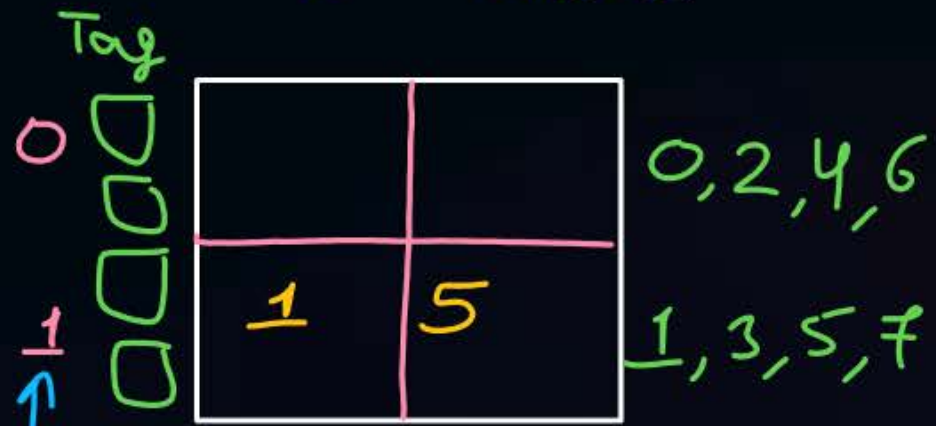
mm block



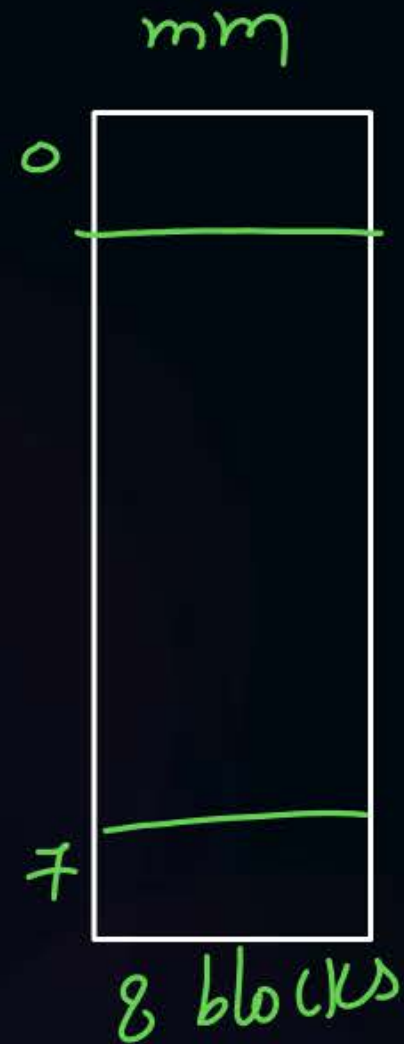


Topic : Set Associative Mapping

2-way set associative
Cache



each index
can have 2
different main
memory blocks



cpu Request mm block:-

1, 5, 1, 5, 2, 5, 1, 5, 1, 5
miss hit



Topic : Set Associative Mapping

$$\text{no. of sets in cache} = \frac{\text{no. of blocks in cache}}{\text{associativity}}$$

$$\text{cm set no.} = (\text{mm block no.}) \% \text{ no. of sets in cache}$$

$$\text{Tag} = \left\lfloor \frac{\text{mm block no.}}{\text{no. of sets in cache}} \right\rfloor$$

mm add.

Tag	cm set no.	byte no.
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$$\text{no. of bits in cm set no.} = \log_2(\text{no. of sets in cache})$$

$$\text{Tag directory size} = \text{no. of blocks in cache} * (\text{Tag} + \text{extra bits})$$

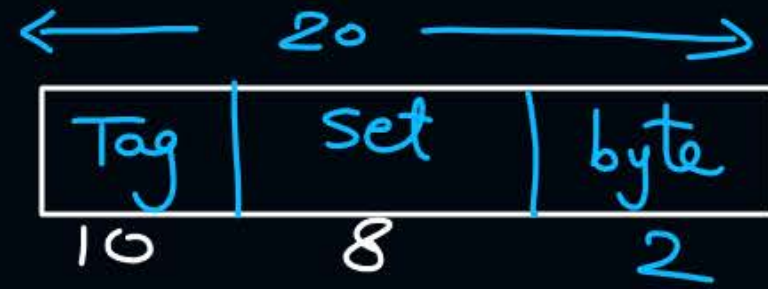
Ques)

mem add. = 20 bits

Cache size = 2K bytes

block size = 4 B = 2^2 B

2-way set associative cache



$$\text{no. of blocks in cache} = \frac{2K \cancel{B}}{4 \cancel{B}} = \frac{2^{11}}{2^2} = 2^9$$

$$\text{no. of sets in cache} = \frac{2^9}{2} = 2^8$$

$\Downarrow$
set no = 8 bits

$$\text{Tag directory size} = 2^9 * 10 \text{ bits}$$

#Q. A computer has a 512Kbyte, 4-way set associative, write back data cache with block size of 16 Bytes. The processor sends 34 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit

1. The number of bits in the tag field of an address is 17 bits
2. The size of the cache tag directory is $2^{15} * (17 + 2 + 1 + 1) \text{ bits} = 2^{15} * 21 \text{ bits}$

34 bits		
Tag	set	byte
17	13	4

$$\text{no. of blocks in cache} = \frac{512 \text{ KB}}{16 \text{ B}} = \frac{2^9 \cdot 2^{10}}{2^4} = 2^{15}$$

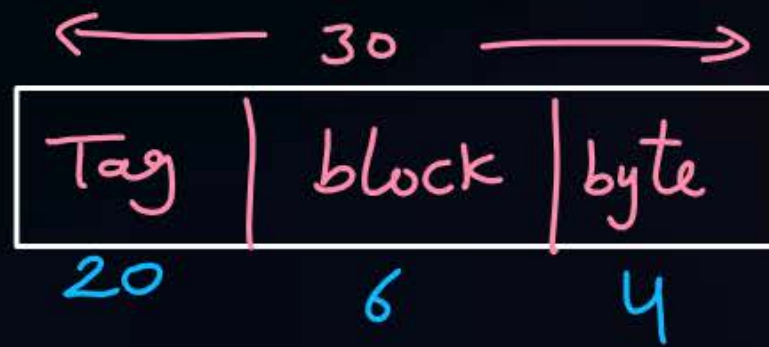
$$\text{no. of sets in cache} = \frac{2^{15}}{4} = 2^{13} \Rightarrow \text{set no.} = 13 \text{ bits}$$



Topic : Tag and Index in Associative Mapping

ex:- block size = 16 B
cm size = 1 KB

Direct mapping:-



← $\log 1K = 10$ →

$$\text{no. of blocks} = \frac{1KB}{16B} = 64 = 2^6$$

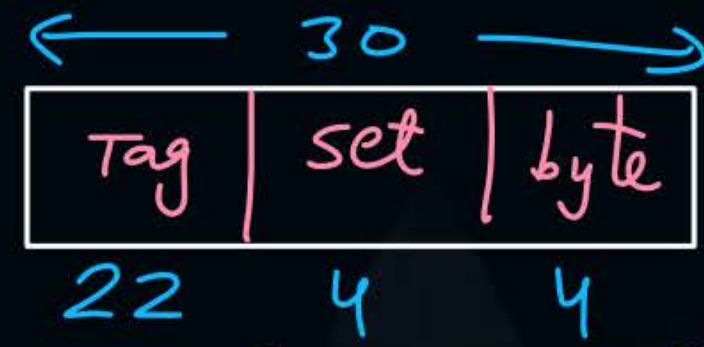
2-way:-



← $\log 1K - \log 2$ →

$$\text{no. of sets} = \frac{2^6}{2} = 2^5$$

4-way



← $\log 1K - \log 4 = 10 - 2 = 8$ →

$$\text{no. of sets} = \frac{2^6}{4} = 2^4$$

← mm add →

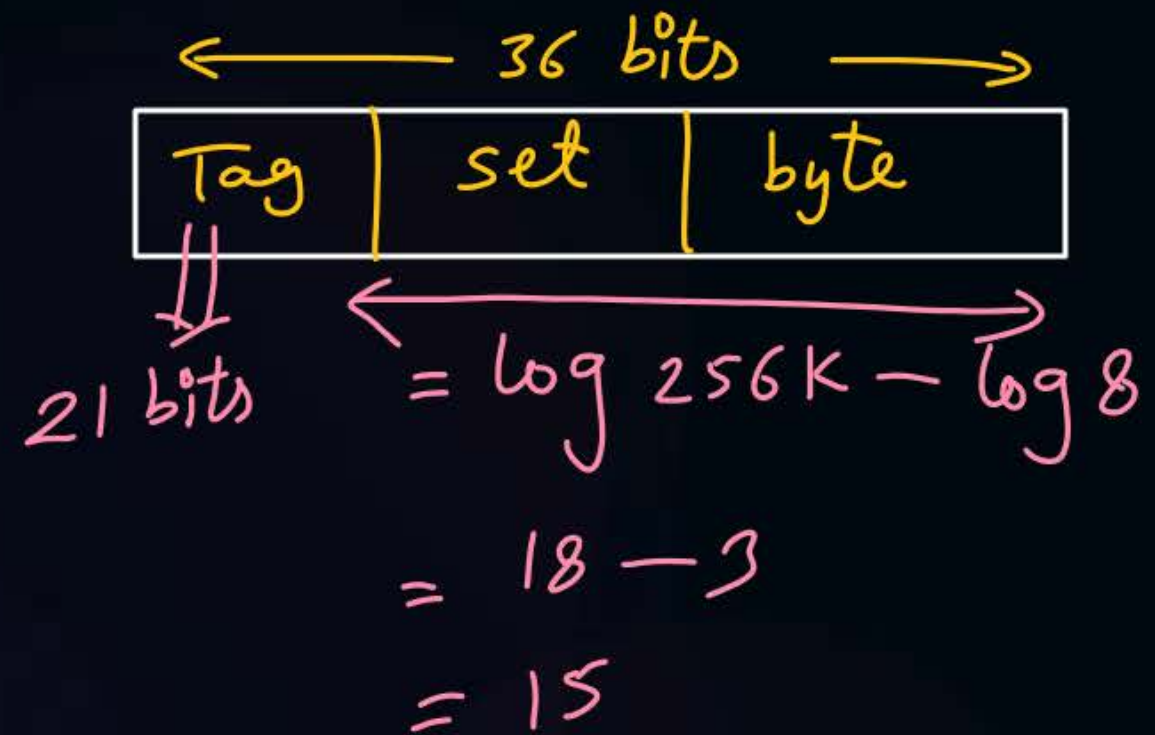


← $\log(\text{cm size}) - \log(\text{associativity})$ →

$$\text{Tag} = \text{mm add.} - \left[\log(\text{cm size}) - \log(\text{associativity}) \right]$$

$$= \text{mm add.} - \log \text{ cm size} + \log \text{ associativity}$$

#Q. The width of the physical address on a machine is 36 bits. The width of the tag field in a 256 KB 8-way set associative cache is 21 bits ?





2 mins Summary



Topic

Cache Initialization

Topic

Set Associative Mapping



Happy Learning

THANK - YOU

